

Child Safety and Activity Monitoring System Using IoT

Kadhiresh S

Department of Computer science and Engineering
Karpagam College of Engineering
Coimbatore-641032 , Tamilnadu
Email: kadhiresh02@gmail.com

Saran kumar C

Department of Computer science and Engineering
Karpagam College of Engineering
Coimbatore-641032 , Tamilnadu
Email: sarankongu23@gmail.com

Tamizhan M

Department of Computer science and Engineering
Karpagam College of Engineering
Coimbatore-641032 , Tamilnadu
Email: tamilanmurugan071@gmail.com

Vidya prabha C V

Department of Computer science and Engineering
Karpagam College of Engineering
Coimbatore-641032 , Tamilnadu
Email: vidyaprabhacvv@gmail.com

Abstract— Child safety and tracking are pressing concerns that garner widespread attention due to the escalating incidence of crimes against children. In response to this, an innovative Internet of Things (IoT) device has been developed to ensure the safety and monitoring of children. This monitoring system aids parents in tracking and overseeing their children's activities, catering to those who seek to maintain constant awareness of their child's whereabouts. It utilizes GPS and telephony services within an Android application to accurately pinpoint the child's location, allowing parents to discreetly monitor messages and track their child without their knowledge. Notably, the application operates discreetly in the background of the child's phone, even after rebooting.

The system, coded in embedded C, interfaces with various sensors including temperature, heartbeat, and tactile sensors, as well as GPS, GSM, and a digital camera. Its standout feature lies in its automated module system, which promptly notifies parents or caregivers via real-time SMS alerts. This proactive approach ensures swift action during emergency situations, emphasizing the importance of continuous monitoring. Additionally, factors such as the child's tactile sensations, body temperature, and heart rate are considered, with parametric analysis facilitating visualization of the obtained results.

Keywords: Child safety system, Child surveillance system, Internet of Things (IoT) technology, IoT gadget, Intelligent wristband.

I. INTRODUCTION:

The term "Internet of Things" (IoT) encompasses a network of interconnected systems and devices equipped with sensors and actuators, all linked to the internet. This network possesses the ability to autonomously perceive and respond to its environment without human intervention. In our study, we propose the implementation of an IoT-enabled smart band tailored for parents to remotely monitor and ensure their child's well-being regardless of their physical presence. This innovative smart band serves as a crucial tool in enhancing child safety and contributes to lowering crime rates by facilitating prompt actions in case of potential dangers. Unlike existing solutions that prioritize other aspects over child security, our proposed system focuses on gathering comprehensive

data to accurately evaluate the child's situation. The integration of IoT technology into this device is motivated by the urgent need for a robust child security system, particularly in Malaysia, where instances of crimes involving children are on the rise.

To address this pressing issue, our system aims to achieve the following objectives:

- Enable real-time tracking of the child's location and collect vital data such as temperature, pulse rate, respiratory rate, and sleep quality.
- Present the collected data alongside established reference values for easy interpretation.
- Trigger notifications if the child ventures outside predefined boundaries or if certain conditions or situations are detected by the device.
- Activate an alarm and initiate video recording upon pressing the emergency button. The system will promptly notify parents via their mobile applications and display real-time video footage.
- Develop a prototype of the smart band that seamlessly interfaces with parents' mobile applications, empowering them to monitor their child's well-being from anywhere.

II. PROPOSED SYSTEM:

A wearable smart band specially crafted for children integrates IoT technology, marking a significant advancement in child safety. This study proposes an approach to enhance child safety, aiming to deliver informative and neutral content tailored for a knowledgeable audience in a general context. The smart band is designed to withstand water submersion, rechargeable, and encompasses a range of features. It incorporates heart rate sensors to measure pulse rate and beats per minute (BPM), along with a sleep quality sensor to assess children's sleep patterns. An altimeter monitors cycling activities and various positions, detecting altitude changes and identifying descents. Complementing this, a pedometer tracks the number of steps taken, while a motion sensor verifies the child's presence.

For comprehensive health monitoring, a sensor measures blood pressure levels and respiratory rates, identifying breathing patterns and rates of inhalation and exhalation. Additionally, a temperature sensor detects body temperature, with a predefined threshold value of 38°C. The adept child support program comprises two modules, showcasing their contrasting functionalities. Leveraging components such as sound sensors and heartbeat sensors, coupled with effective tactics and operational strategies, the system executes tasks proficiently. The SMS alert mechanism is facilitated through a web server, ensuring continuous device operation. Employing a server-based approach, a concise layout elucidates the process of sending SMS messages.

A. WORKING

The child safety and activity monitoring system leverage IoT devices, encompassing wearable sensors and environmental detectors, to monitor a child's vital signs, movements, and surroundings. These wearables, equipped with GPS and communication modules, transmit data to a central processing unit, which utilizes algorithms to examine the information for irregularities. Visual monitoring is facilitated by strategically placed cameras, while a cloud platform securely stores and processes the data. Parents gain access to real-time updates and location tracking via an intuitive mobile app or web interface. The system issues alert for unusual activities or emergencies, prompting predefined responses and ensuring swift parental action. Emphasizing privacy and security, the system complies with data protection regulations and ethical standards. Through integrated technology, the goal is to deliver a comprehensive child safety solution with continuous monitoring and rapid emergency response capabilities. Implementation of low-power IoT devices and sensors ensures prolonged battery life.

III. BLOCK DIAGRAM:

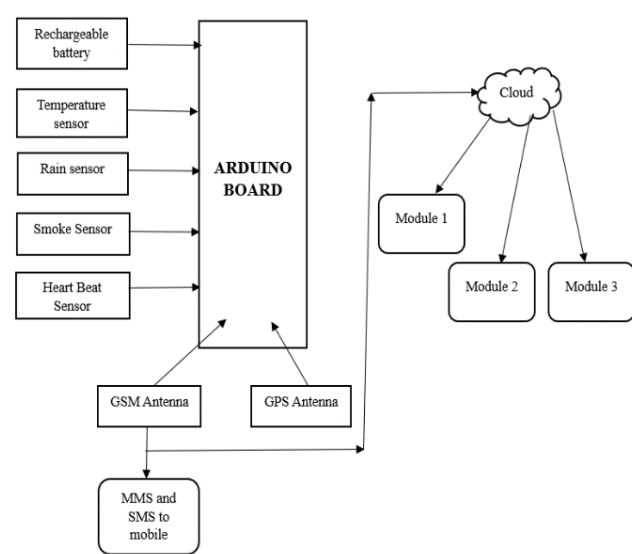


Fig 1. BLOCK DIAGRAM

IV. HARDWARE DESCRIPTION

A. ARDUINO BOARD

A digital representation of the project's identity is manifested through an electronic development panel. This panel features an open-source electronic circuitry, housing a microcontroller controllable via a PC, aimed at fostering utilization. Smart hardware assumes a crucial role in interdisciplinary endeavours. Arduino serves as a primary tool in crafting intelligent technologies. Ventures and endeavours are underway to craft distinctive environmental sensors capable of detecting diverse factors, including temperature and wind. Light and weight exhibit an intrinsic connection. Arduino seamlessly integrates with a myriad of computer projects and operates on an open-source programming language platform.



Fig 2. ARDUINO BOARD

B. TEMPERATURE SENSOR

The DHT11 Temperature and Humidity Sensor is equipped with a calibrated digital signal output, facilitating temperature and humidity sensing capabilities. Integrated with a high-performance 8-bit microcontroller, its advanced technology guarantees superior reliability and long-term stability. This sensor incorporates both a resistive element and a wet NTC temperature measuring device, boasting outstanding quality, rapid response time, anti-interference capabilities, and optimal performance.

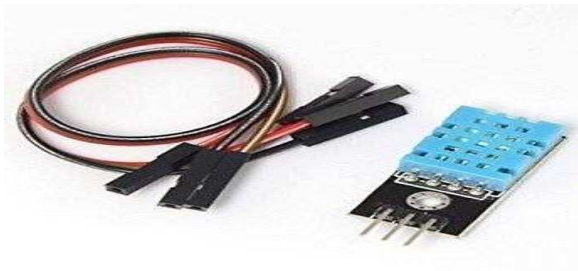


Fig 3. TEMPERATURE SENSOR

C. GAS SENSOR

The MQ-135 Gas sensors find application in air quality monitoring devices and are capable of detecting or measuring NH₃, NO_x, Alcohol, Benzene, Smoke, and CO₂. The MQ135 sensor module is equipped with a Digital Pin, allowing standalone operation without a microcontroller, which proves convenient when targeting specific gas detection. For measuring gases in parts per million (PPM), the analog pin must be utilized. Operating at 5V TTL levels, the analog pin is compatible with most standard microcontrollers.



Fig 4. GAS SENSOR

D. RAIN SENSOR

A water sensor functions as a type of switch, designed to detect rainfall or water flow. Operating on a principle akin to a switch, its functionality entails a normally closed state, which opens upon detection of rain or water presence.



Fig 5. RAIN SENSOR

E. WIFI MODULE

The ESP8266 device is an affordable and user-friendly option for adding internet connectivity to your projects. It offers dual functionality as both an Access Point (capable of creating a hotspot) and a Station (able to connect to Wi-Fi), simplifying data retrieval and internet uploading for Internet of Things applications. Additionally, it can access data from the internet using APIs, enhancing project capabilities significantly. Furthermore, it supports programming via the Arduino IDE, enhancing its ease of use. However, this version of the module only features 2 GPIO pins (expandable to 4 with hacks), necessitating its use alongside another microcontroller like Arduino. Alternatively, standalone versions such as ESP-12 or ESP-32 offer more GPIO pins. If you seek a module to kickstart your IoT journey or provide internet connectivity to your project, the ESP8266 is an ideal choice.

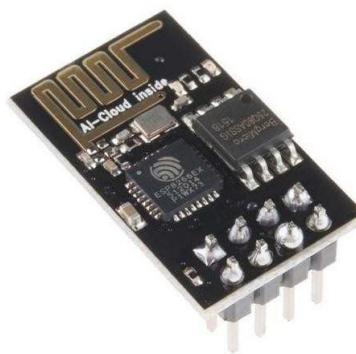


Fig 6. WIFI MODULE

F. GSM CONNECTION UNIT

GSM, an abbreviation for Global System for Mobile Communication, serves as the foundation for a vast majority of mobile telecommunications globally. Renowned as one of the most successful wireless technologies, GSM has garnered widespread recognition and acclaim. This thriving and continually evolving platform offers a wide array of multimedia and voice services to users. The existing GSM network adheres to established standards, ensuring reliability and compatibility. For this project, the GSM module employed is the GSM sim800L, boasting worldwide applicability and specifications. Its functionalities encompass sending and receiving phone calls and text messages, as well as providing access to various additional features.

V. SOFTWARE REQUIREMENTS:

A. EMBEDDED C

Embedded C, an extension of the widely-used C programming language, was introduced by the C Standards Committee to address the unique challenges of programming embedded systems. Initially, assembly language dominated the field of embedded systems programming. However, C emerged as a dominant force due to its portability and simplified development process. In the early stages, developers encountered limitations with available tools, and C emerged as a superior alternative, providing reliability, scalability, and cross-platform compatibility. Unlike assembly language, which offered efficiency but posed challenges for larger projects, C boasted advantages such as a shallower learning curve, processor independence, and streamlined direct hardware control. With the widespread availability of compilers, C swiftly became the preferred choice, offering flexibility and robust support for low-level data manipulation.

VI. RELATED WORK:

A. INTERNET OF THINGS (IoT)

The Internet of Things (IoT) encompasses the interconnection of objects via networks, endowing them with intelligence and facilitating communication. Through IoT, objects can utilize internet connectivity to exchange data, interact, and make decisions across various locations. This paradigm introduces hyperconnectivity, enabling seamless communication among individuals and organizations. Regarded as a transformative revolution,

IoT impacts information technology, human lifestyles, and business operations. Its advancements empower organizations to streamline and monitor business processes efficiently. In society, IoT simplifies daily routines and enhances quality of life through innovations like smart homes, smart cities, and various devices and applications. However, security and privacy concerns persist as IoT gathers potentially sensitive data, necessitating robust safeguards.

B. SENSOR

A sensor serves as a tool for measuring values and translating them into data. A prevalent example is the temperature sensor, employed to gauge the heat of an object. Another variant, the proximity sensor, is utilized for object detection purposes. To measure pressure, a pressure sensor computes the applied pressure. An optical sensor is capable of discerning the intensity of light. Furthermore, a humidity sensor identifies the presence of water vapor in the air. Finally, a microsensor is engineered to gather and transmit environmental information.

C. SAFETY DEVICE

A safety device functions as a crucial tool in safeguarding individuals from potential harm and peril. In a study led by [1], researchers proposed a child safety wearable device powered by the Raspberry Pi 3. This intelligent device gathers data from various sources such as the Pi camera, pulse sensor, and sound sensors. The collected data is then relayed to parents' smartphones via SMS using a GSM shield. Furthermore, parents receive real-time images captured by the Pi camera alongside location information of their children through GPS tracking. Another research endeavour focused on the development of a smartwatch tailored specifically for women's security. This innovative watch incorporates sensors that monitor the wearer's heartbeat rate. In the event of any attacks or emergencies where the heartbeat rate escalates significantly, an alarm is triggered. Subsequently, pre-registered contacts and even the nearest police station are automatically alerted through phone calls. Leveraging GPS technology, authorities can promptly respond to emergencies by pinpointing the location.

D. RESULT AND DISCUSSION:

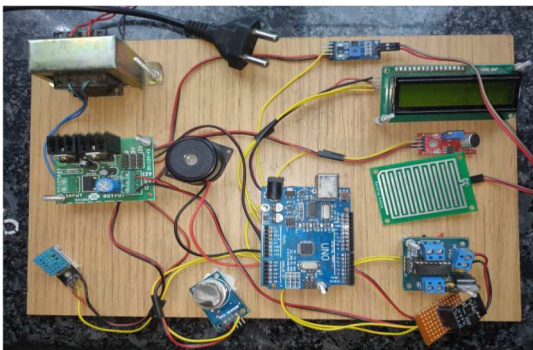


FIG:HARDWARE VIEW

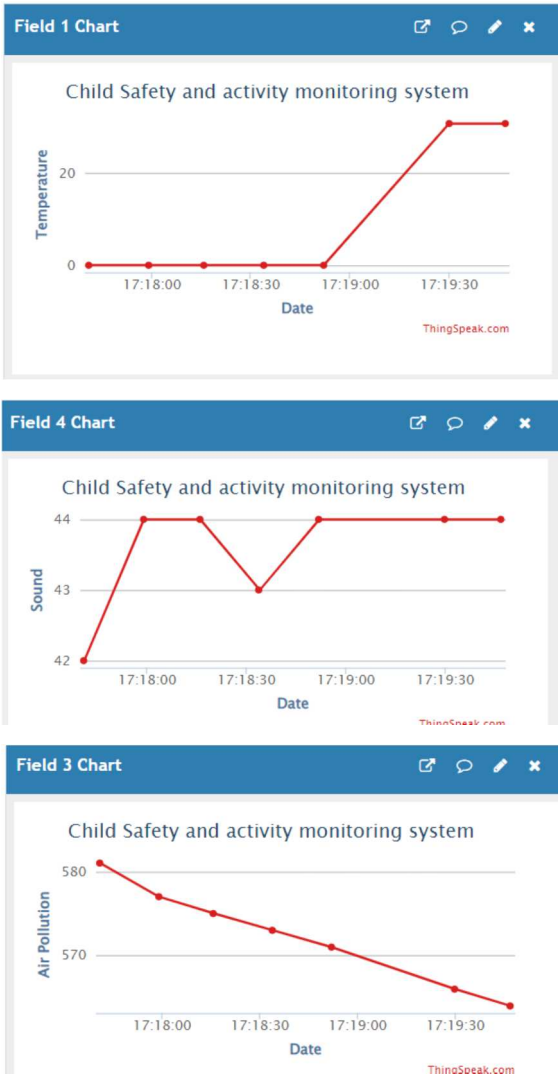


Fig 8. SAMPLE OUTPUT

VII. CONCLUSION:

The research provides in-depth insights into the subject matter, offering lucid explanations on the Internet of Things (IoT), child safety concerns, and the imperative need for implementation. Previous research has delved into child security systems, aiming to develop a smart system grounded in IoT principles. This system facilitates remote monitoring of children by parents, with notifications promptly dispatched in case of emergencies. Effective communication with parents is vital for prompt action in safeguarding their children, thereby contributing to a reduction in crime rates. However, the proposed device currently lacks the requisite durability and reliability, along with the essential functionality to operate as a mobile phone. Hence, future enhancements will focus on expanding the array of features, incorporating improved software, applications, and hardware components into the system. The aim is to bolster the system's functionality and efficacy, ensuring the safety of children through intelligent means.

REFERENCES:

- [1] Libguides.wits.ac.za. 2020. Libguides: Research Support: Research Methodology.
- [2] AbdelRahman H. Hussein. Internet of Things (IOT): Research Challenges and Future Applications. (IJACSA) International Journal of Advanced Computer Science and Applications. 10(6). 2019.
- [3] L. Tan, K. Yu, F. Ming, X. Cheng, G. Srivastava, "Secure and Resilient Artificial Intelligence of Things: a HoneyNet Approach for Threat Detection and Situational Awareness", IEEE Consumer Electronics Magazine, 2021
- [4] H. Li, K. Yu, B. Liu, C. Feng, Z. Qin and G. Srivastava, "An Efficient Ciphertext-Policy Weighted Attribute-Based Encryption for the Internet of Health Things," IEEE Journal of Biomedical and Health Informatics, 2021
- [5] K. Yu, Z. Guo, Y. Shen, W. Wang, J. C. Lin, T. Sato, "Secure Artificial Intelligence of Things for Implicit Group Recommendations", IEEE Internet of Things Journal, 2021
- [6] Nishit Raghuwanshi Rudra. 8 Best GPS Trackers For Kids In 2020 To Ensure They Are Safe. 2020.
- [7] Hanan M. Shukur, Lailan M. Haji, Mohammad A. M. Sadeeq, Omar M. Ahmed, Rizgar R. Zebari and Shakir M. Abas. Journal of Applied Science and Technology Trends. 1(2). (2019).
- [8] Dissertation.laerd.com. Quota Sampling | Lærd Dissertation. 2021.
- [9] Dr. R. Nagaraja and P. Elamathi. Smart Children Safety Using Wearable Device - A Review. International Journal of Advanced Research in Electrical, Electronics and Instrumentation Engineering. 8(11). 2019.
- [10] Dr. J. Jegathesh Amalraj, J. Jereena John and S. Banumathi. IOT Sensors And Applications: A Survey. International Journal Of Scientific & Technology Research. 8(8). 2019
- [11] AbdelRahman H. Hussein. Internet of Things (IOT): Research Challenges and Future Applications. (IJACSA) International Journal of Advanced Computer Science and Applications. 10(6). 2019.
- [12] Ali, Mohammed Hasan, and Qasim Al Azze, "Design and implementation a security system for bank using voice recognition," International Journal of Power Electronics and Drive Systems (IJPEDS), vol. 10, no. 4, pp. 2126-2129, 2019.
- [13] Abdullah, Ali Najim, and Mohammed Hassan Ali, "Direct torque control of IM using PID controller," International Journal of Electrical and Computer Engineering (IJECE), vol. 10, no 1, pp. 617-625, 2020.
- [14] V Adewopo, N Elsayed, K Anderson - SoutheastCon 2023, 2023
- [15] Dr. J. Jegathesh Amalraj, J. Jereena John and S. Banumathi. IOT Sensors And Applications: A Survey. International Journal Of Scientific & Technology Research. 8(8). 2019